

Magnetic Actuation System Dataset Generation and Characterization

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August 2024

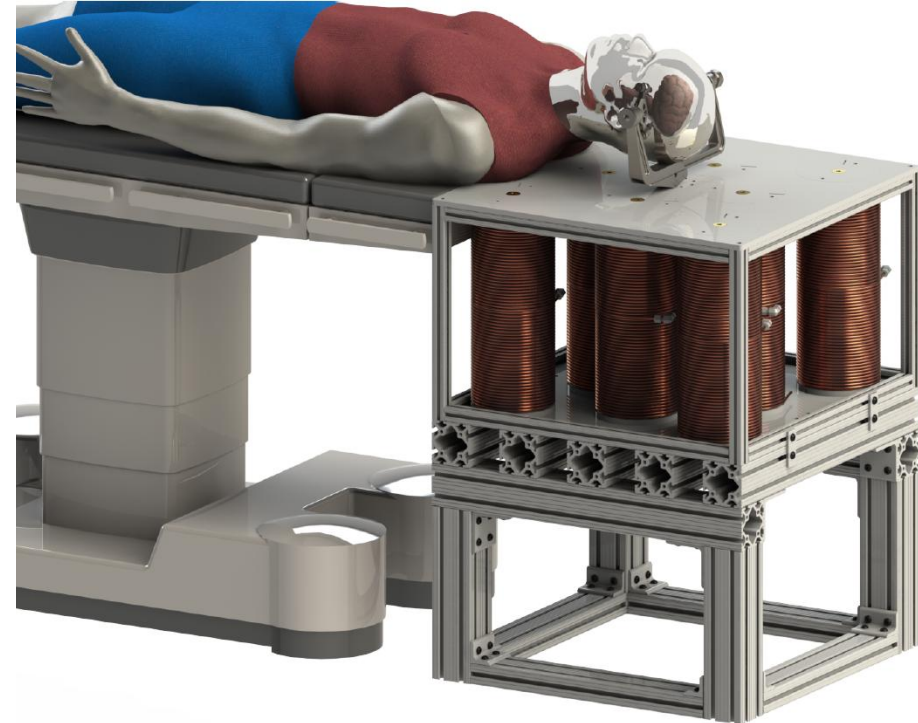
Outline

- Introduction
- Objectives
- Approach
- Result and discussion
- Conclusion and Future work



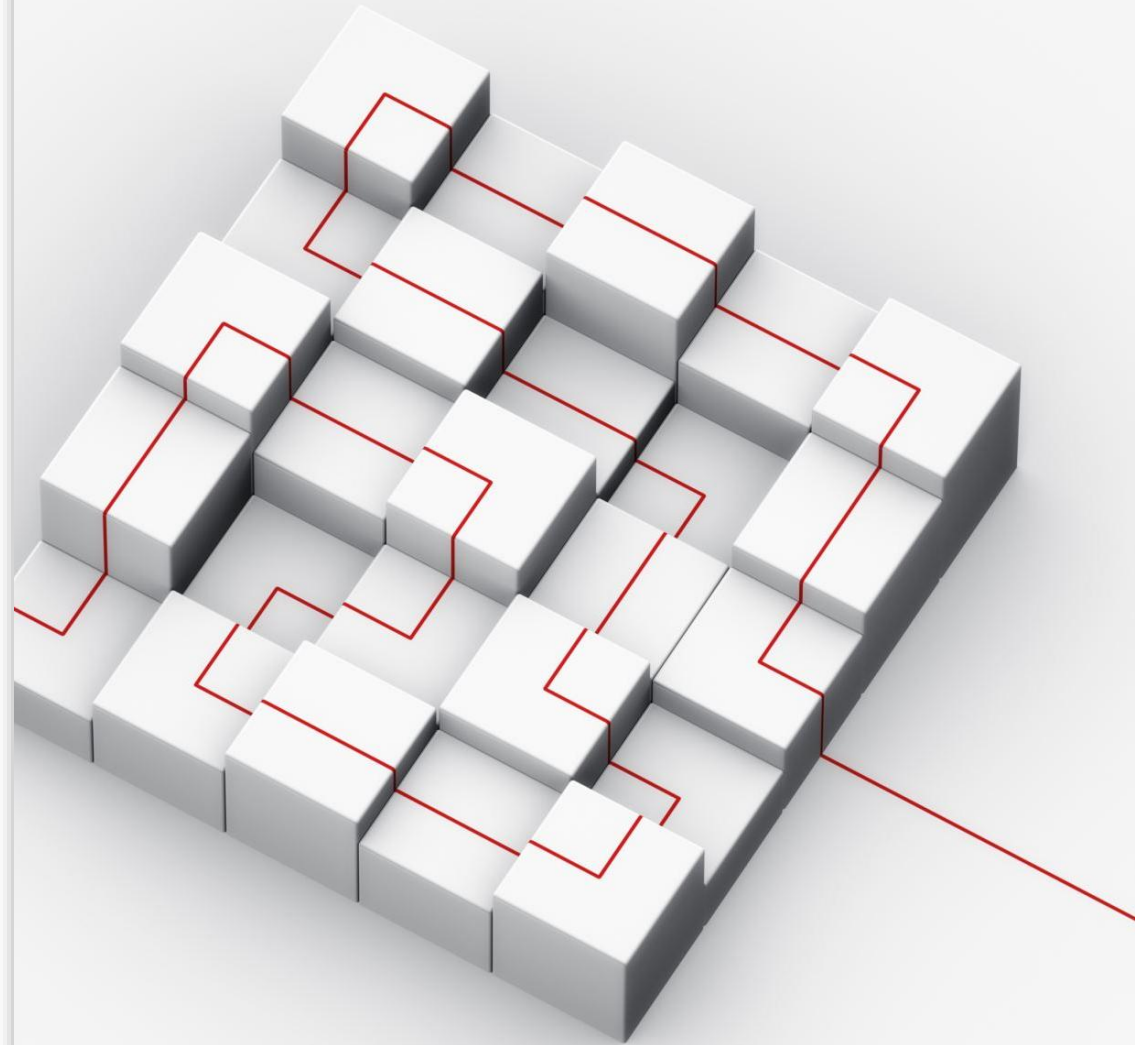
Introduction

- Surgical tools are required to operate in different positions and orientations to access their target point
- Currently, the coil table has been calibrated only for a single point.
- The 3D magnetic field generated needs to be known to expand capability to 3D control.



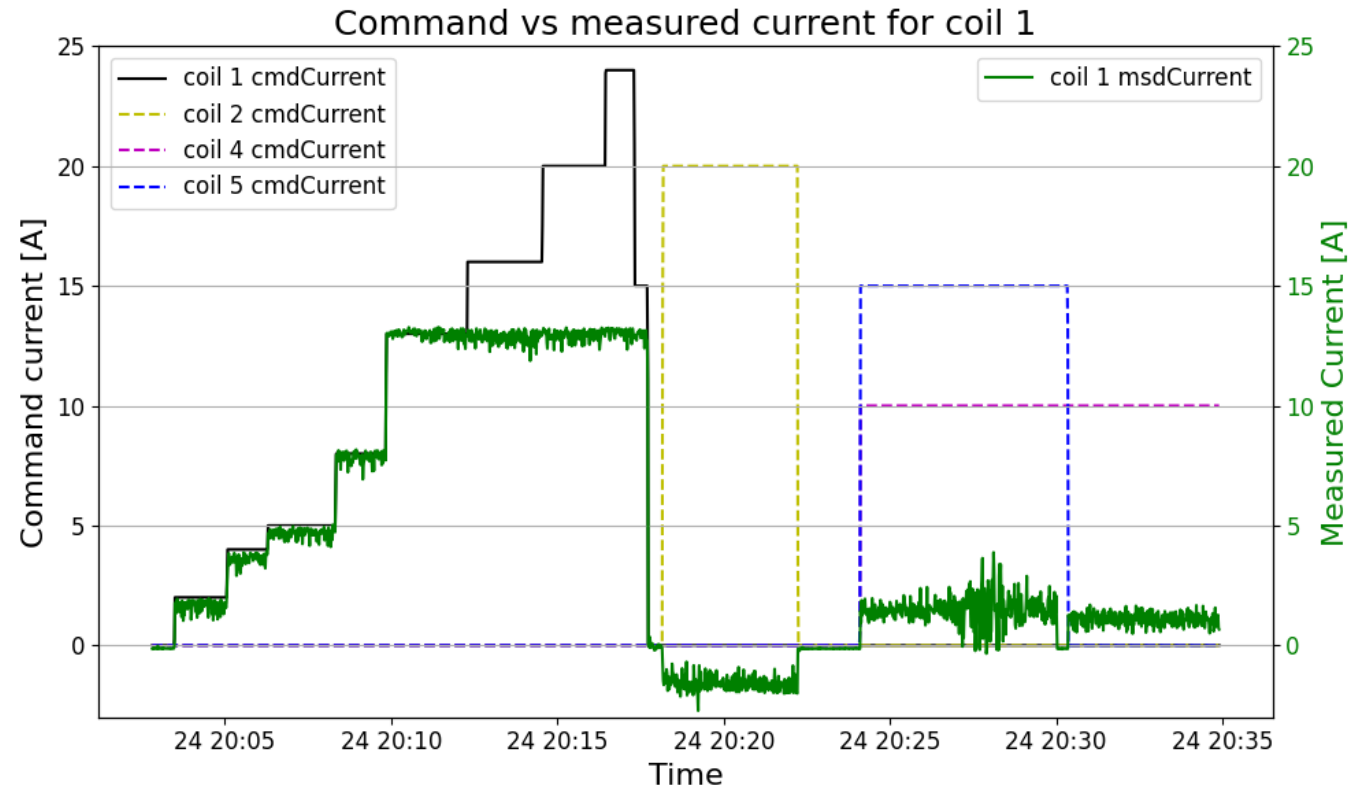
Objectives

- Primary Objective
 - Generate a 3D magnetic field measurement dataset for training a machine learning model.
- Secondary Objectives
 - Validate the existing model (single pt.).
 - Determine the bias of the dipole model for the 3D space around the system.



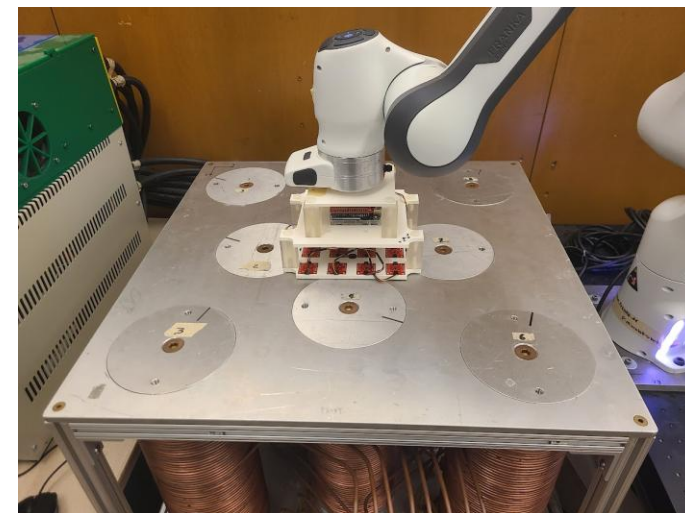
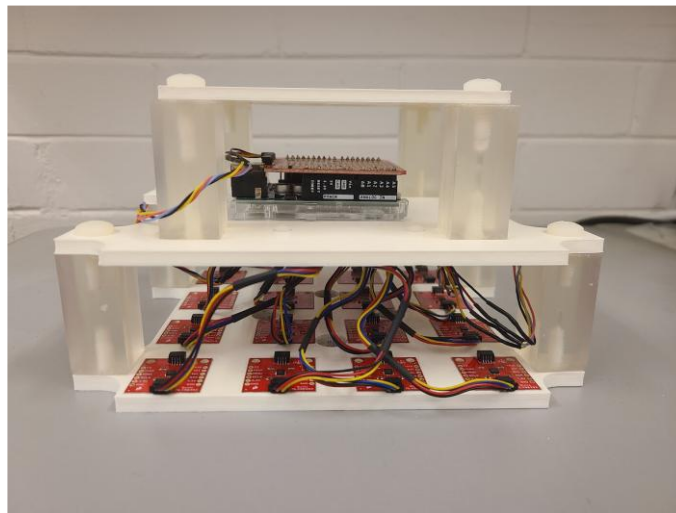
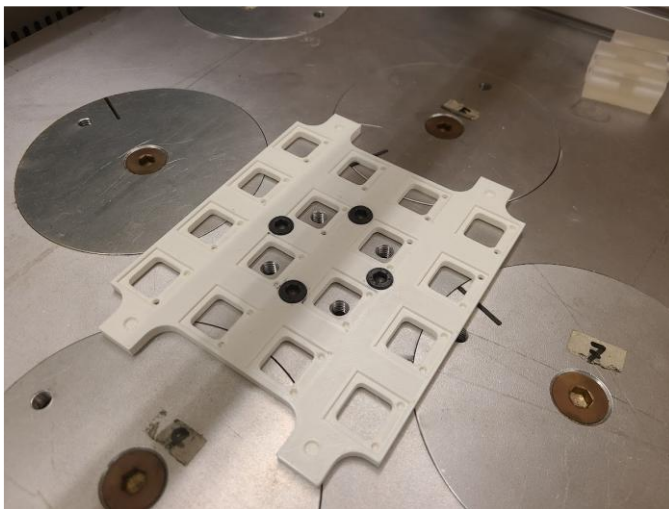
Approach

- For 8 input current combination
 - Every combination
 - Monte Carlo Method
 - Super-positioning – only valid in linear range [Kummer et al.]
- Max current is found to be limited to $\pm 13\text{A}$
 - $\pm 15\text{A}$ is considered for measuring range
- Operation is in the linear range
 - the generated magnetic field is linearly correlated with the input current.
 - Coarse discretization of 2A is chosen
- For 8 coils, over 2.5 billion sweeps of the 3D space is required!
- Method of Superposition adopted
 - Only 120 sweeps needed!



Sensor array

Sensor model: Melexis MLX90393 from sparkfun



Sensor Verification

Test 1: z-axis Fields

Test 2: x-axis Fields

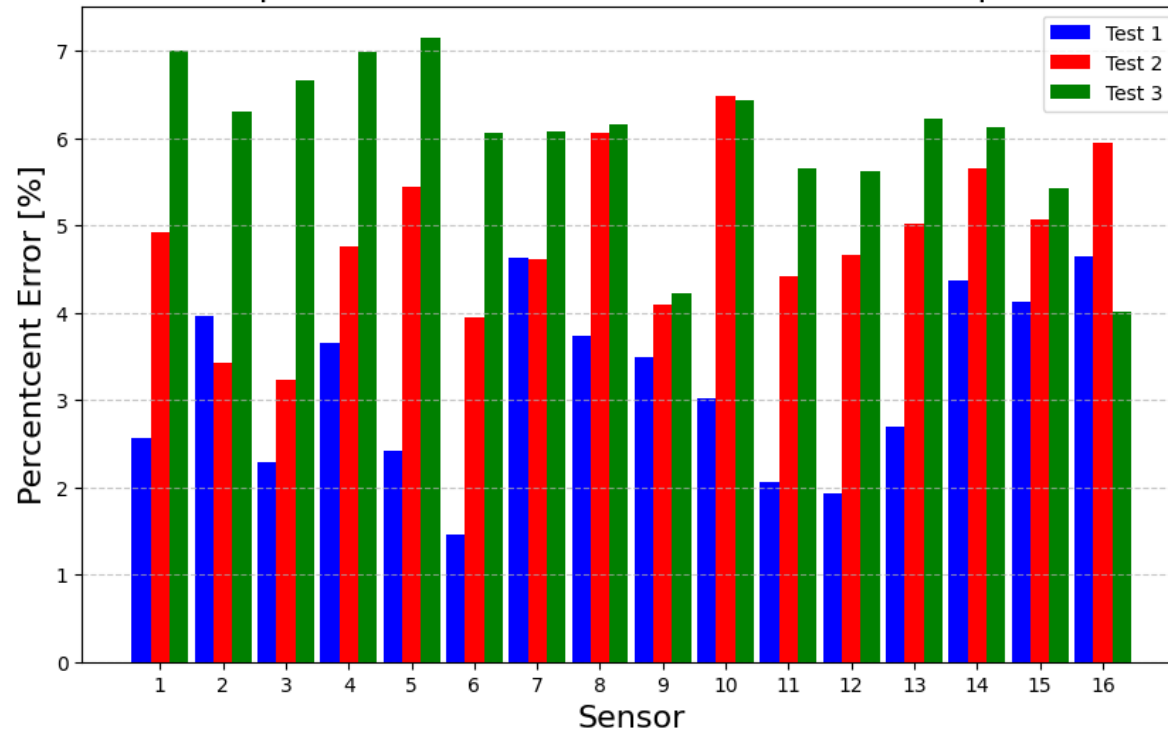
Test 3: y-axis Fields



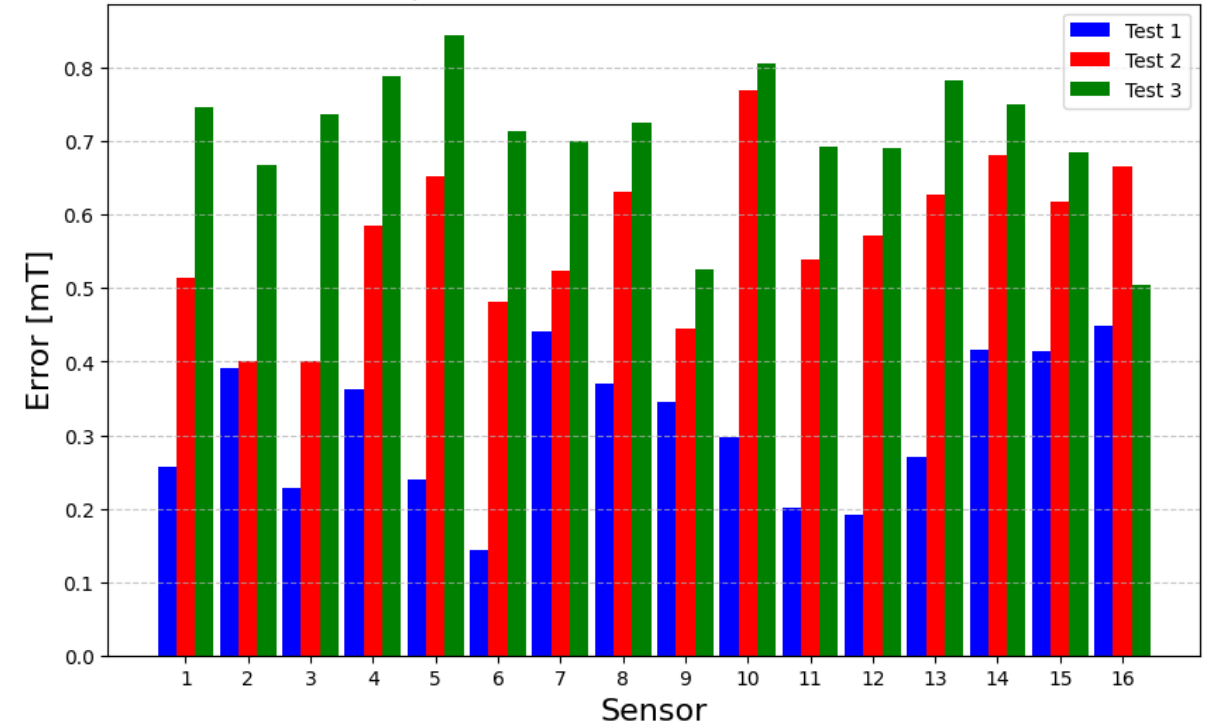
Vs



Gauss-probe vs MLX90393 Measurement Discrepancies



Gauss probe vs MLX90393 measurement



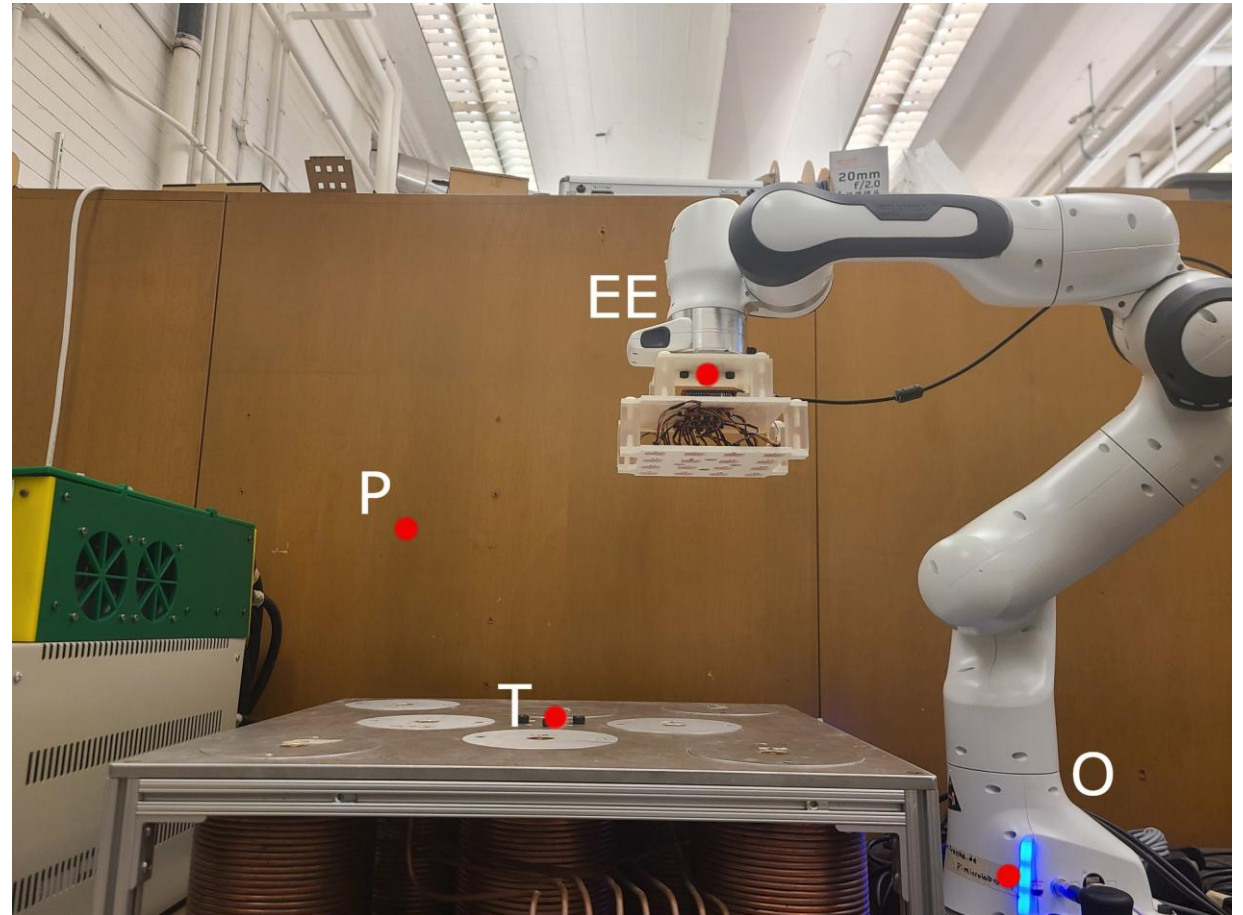
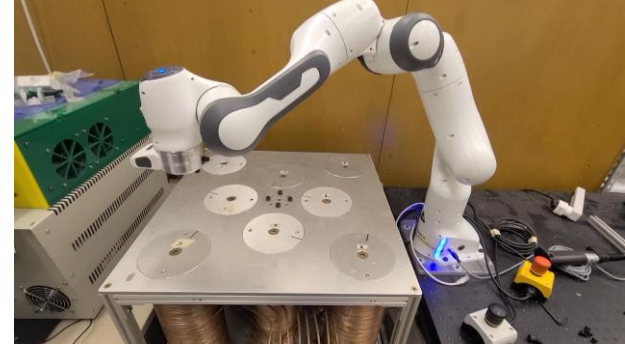
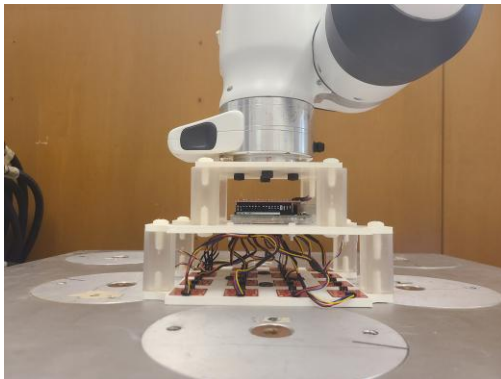
Robot Control

$$T_{TO} = T_{TE}T_{EO}$$

$$T_{OE} = \begin{bmatrix} C_{OE} & r_o^{EO} \\ 0^T & 1 \end{bmatrix}$$

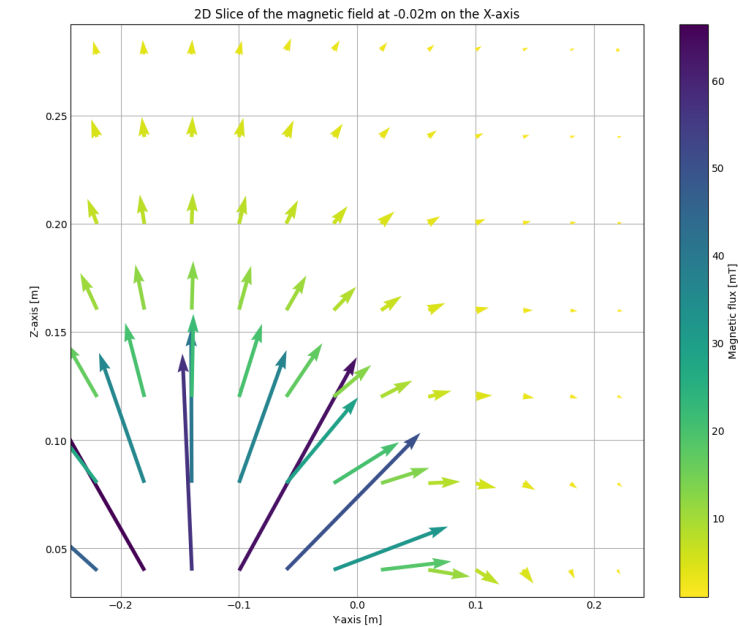
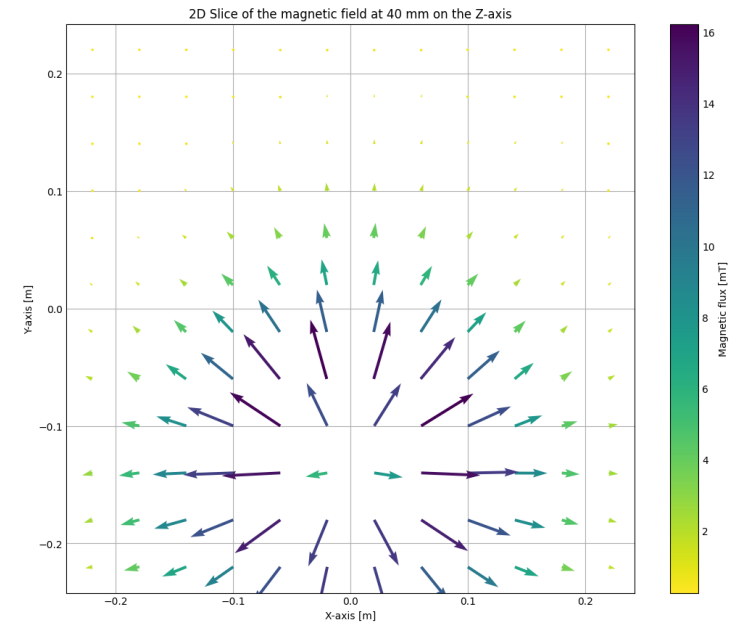
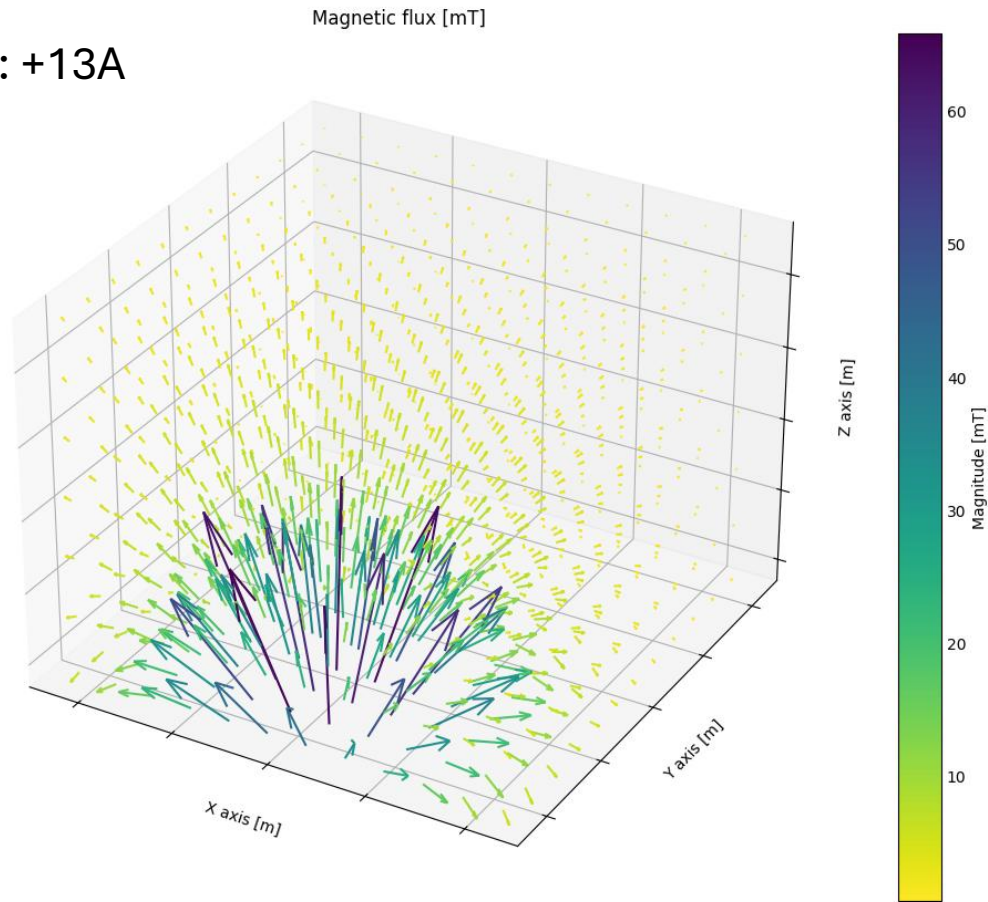
$$T_{EO} = T_{OE}^{-1}$$

$$T_{TE} = \begin{bmatrix} I^{3 \times 3} \cdot C_x(\pi) \cdot C_z(\pi/4) & r_T^{ET} \\ 0^T & 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & -1 & 0.1064 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

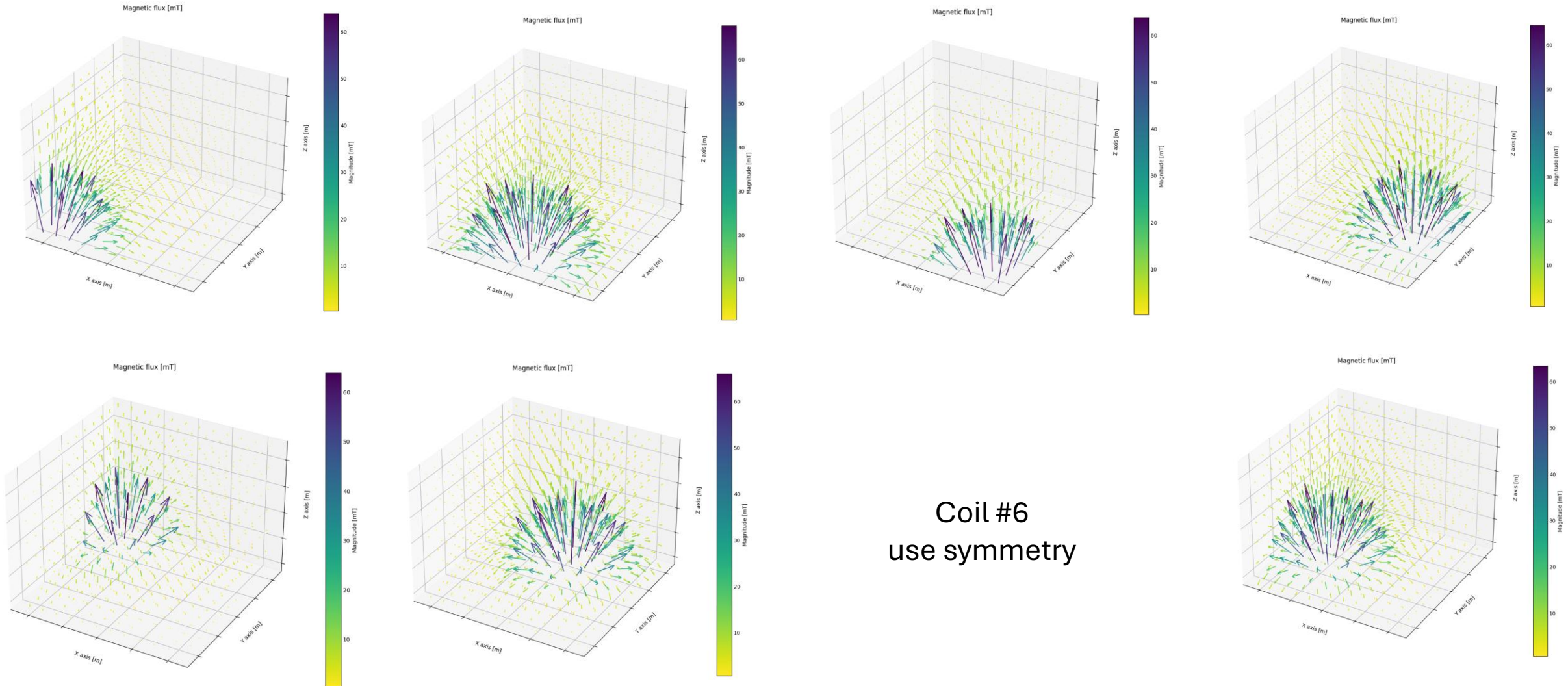


Result and discussion

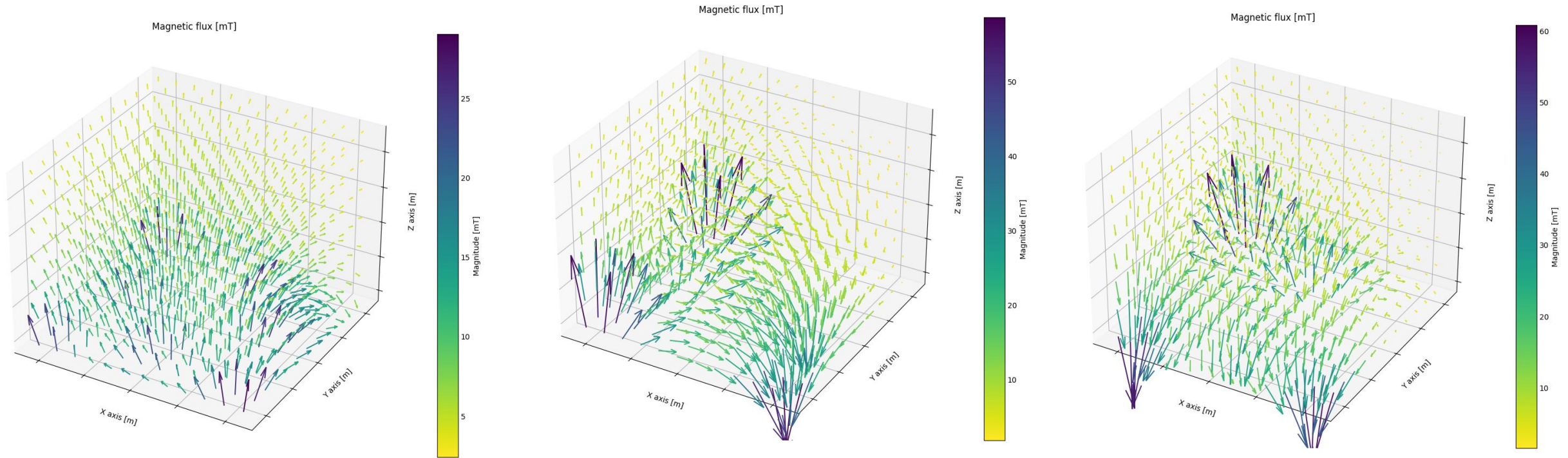
Coil #2 : +13A



Result and discussion

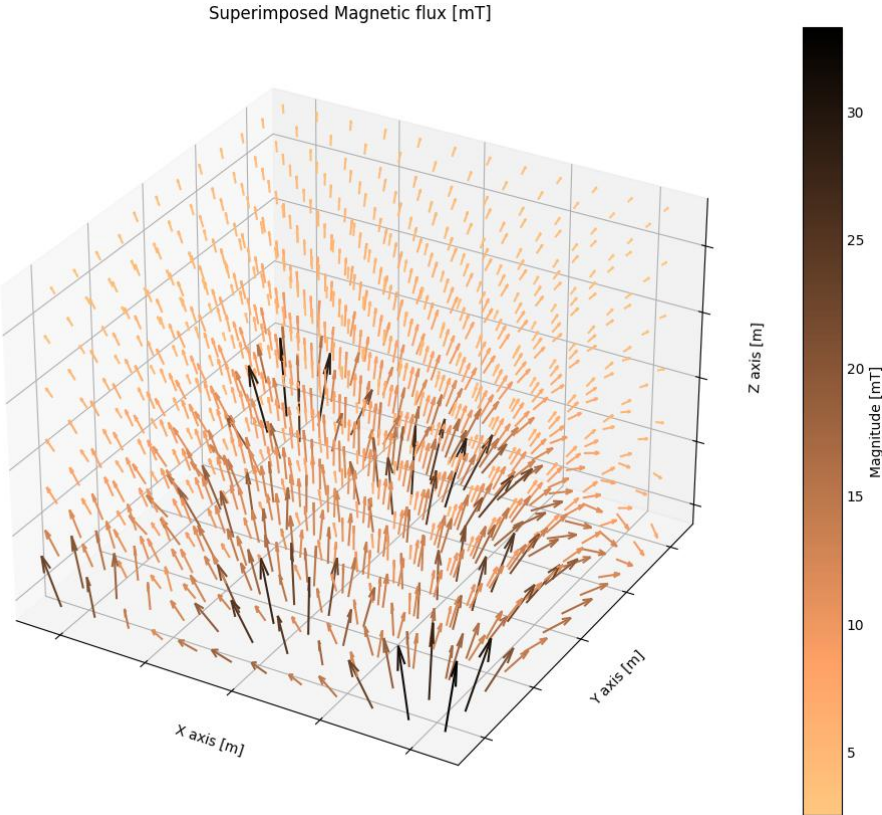


Magnetic fields by multiple coils

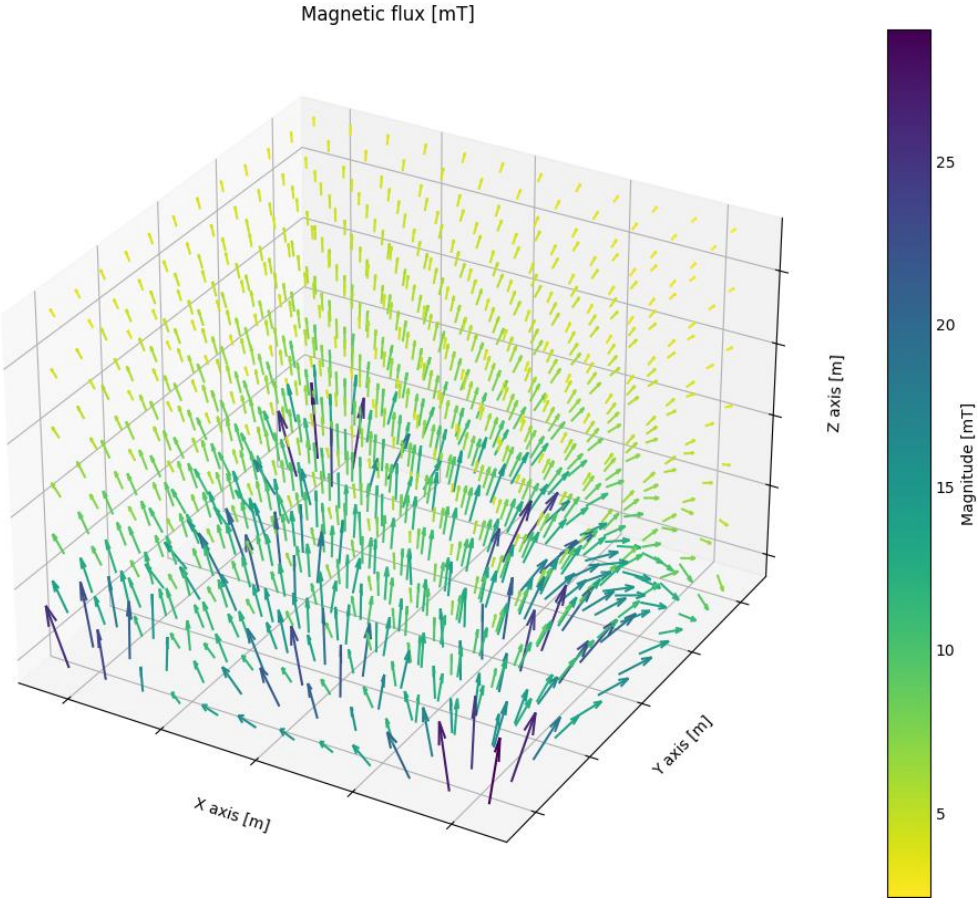


Super positioning

Super-positioned



Measured



mean U_error: 0.061 mT
mean V_error: 0.120 mT
mean W_error: -0.22 mT

	Coil 1	Coil 2	Coil 3	Coil 4	Coil 5	Coil 6	Coil 7	Coil 8
Current setting	5	5	5	5	5	0	5	5

Conclusion and Future work

- The Primary objective of preparing the dataset for model training has been completed.
- Secondary objectives such as calculated bias from the dipole model is on-going.
- Magnetic field gradient is best calculated after a model with fine resolution is trained – Future work.



Thank you!

- Any Questions?

